



INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES)
DHANBAD - 826004, (JHARKHAND), INDIA

मध्यावदी परीक्षा (Mid Semester Examination)

NAME : _____ Adm. No. : _____

SUBJECT : _____ Part : _____ Semester : _____

SIGNATURE OF INVIGILATOR : _____ DATE : _____

Question No.	1	2	3	4	5	6	7	8	9	10	TOTAL
Marks Obtained											

TOTAL MARKS OBTAINED : _____ SIGNATURE OF EXAMINER : _____

Q.3: A set of triaxial test was conducted on rock specimens of the same rock type. The raw data of the test is given as under:

Specimen No.	Average Dia of the Specimen (mm)	Axial load (tonnes)	Confining pressure (Mpa)
1	61.14	2.4	1.4
2	60.81	3.2	2.5
3	61.59	4.2	3.1
4	60.52	5.4	5.2
5	61.23	6.7	5.5

It is assumed that the failure of the rock is governed by the Mohr-Coulomb failure criteria.

- Determine the axial stress on each specimen and tabulate them alongside the confining stress.
- Show that the slope and intercept of the best fit linear regression line on the σ_1 vs. σ_3 graph can be determined without the

following set of equations

$$\begin{pmatrix} m \\ b \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^n \sigma_{3i}^2 & \sum_{i=1}^n \sigma_{3i} \\ \sum_{i=1}^n \sigma_{3i} & n \end{pmatrix}^{-1} \begin{pmatrix} \sum_{i=1}^n \sigma_{3i} \sigma_{1i} \\ \sum_{i=1}^n \sigma_{1i} \end{pmatrix}$$

(C) Determine the cohesion and angle of internal friction of the rock:

Solution:-

For Specimen 1.

$$\text{axial load} = 2.4 \text{ ton.}$$

$$= 2.4 \times 9.8 \times 1000 \text{ N.}$$

$$\text{Area of Specimen} = \frac{\pi \times (61.14)^2}{4} \text{ mm}^2.$$

$$\sigma_{11} \therefore \left. \begin{array}{l} \text{Axial Stress on} \\ \text{the Specimen} \end{array} \right\} = \frac{2.4 \times 9.8 \times 1000}{\frac{\pi \times (61.14)^2}{4}} \text{ N/mm}^2$$

$$= 8.01 \text{ MPa.} \quad [\because 1 \text{ N/mm}^2 = 1 \text{ MPa}]$$

For Specimen 2

$$\sigma_{12} \therefore \left. \begin{array}{l} \text{Axial Stress on} \\ \text{Specimen} \end{array} \right\} = \frac{3.3 \times 9.8 \times 1000}{\frac{\pi \times (60.81)^2}{4}} \text{ N/mm}^2$$

$$= 11.14 \text{ MPa.}$$

For Specimen-3

$$\sigma_{13} = \frac{4.2 \times 9.8 \times 1000}{\pi \times \frac{(61.59)^2}{4}} \text{ N/mm}^2 = 13.82 \text{ MPa.}$$

For Specimen-4

$$\sigma_{14} = \frac{5.4 \times 9.8 \times 1000}{\pi \times \frac{(60.52)^2}{4}} \text{ N/mm}^2 = 18.40 \text{ MPa.}$$

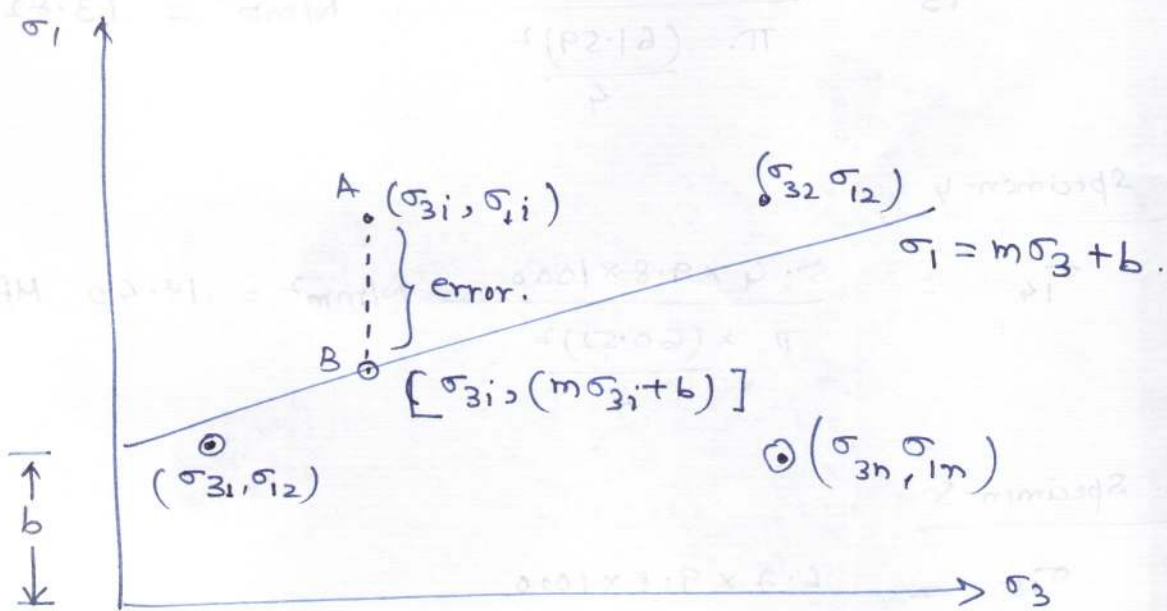
For Specimen-5

$$\sigma_{15} = \frac{6.7 \times 9.8 \times 1000}{\pi \times \frac{(61.23)^2}{4}} \text{ N/mm}^2 = 22.30 \text{ MPa.}$$

Tabulating the calculations we get .

Specimen No	σ_1 (MPa).	σ_3 (MPa)
1	8.01	1.4
2	11.14	2.5
3	13.82	3.1
4	18.40	5.2
5	22.30	5.5

The aforementioned data can be fitted using the generalised linear curve fitting .



The objective is to fit to points $(\sigma_{3i}, \sigma_{1i})$ to the straight line $\sigma_1 = m\sigma_3 + b$.

[m & b are unknown].

The best fit line will have least MSE (Mean Square Error).

For a given value of σ_{3i} the actual value of σ_{1i}

However, for σ_{3i} the value of σ_{1i} on the best fit curve will be } = $m\sigma_{3i} + b$.
given by

$$\therefore \text{Error} = \sigma_{1i} - [m\sigma_{3i} + b] \text{ distance AB.}$$

$$\therefore \text{MSE} = (\text{distance AB})^2.$$

$$\therefore e(m, b) = \text{MSE}.$$

$$\therefore e(m, b) = (\sigma_{3i} - \sigma_{3i})^2 + [\sigma_{1i} - (m\sigma_{3i} + b)]^2$$

$$= \sigma_{1i} - (m\sigma_{3i} + b)^2$$

The total MSE will be

$$E(m, b) = \sum_{i=1}^n [\sigma_{1i} - (m\sigma_{3i} + b)]^2$$

$E(m, b)$ will have extreme value when.

$$\frac{\partial E}{\partial m} = 0 \quad \& \quad \frac{\partial E}{\partial b} = 0$$

$$\therefore \frac{\partial E}{\partial m} = \sum_{i=1}^n 2 [\sigma_{1i} - \{(m\sigma_{3i}) + b\}] \times (-\sigma_{3i}) = 0$$

$$\& \frac{\partial E}{\partial b} = \sum_{i=1}^n 2 [\sigma_{1i} - (m\sigma_{3i} + b)] [-1] = 0$$

$$\text{or, } \left. \begin{aligned} \sum [\sigma_{1i} - (m\sigma_{3i} + b)] \sigma_{3i} &= 0 \\ \sum [\sigma_{1i} - (m\sigma_{3i} + b)] &= 0 \end{aligned} \right\}$$

Expanding the two sets we get

$$\begin{aligned} m \sum \sigma_{3i}^2 + b \sum \sigma_{3i} &= \sum \sigma_{1i} \sigma_{3i} \\ n \sum \sigma_{3i} + b \sum 1 &= \sum \sigma_{1i} \end{aligned}$$

This can be written in matrix form as ,

$$\begin{pmatrix} \sum \sigma_{zi}^2 + \sum \sigma_{zi} \\ \sum \sigma_{zi} + n \end{pmatrix} \begin{pmatrix} m \\ b \end{pmatrix} = \begin{pmatrix} \sum \sigma_{zi} \sigma_{ii} \\ \sum \sigma_{ii} \end{pmatrix}$$

$$\text{or } \begin{pmatrix} m \\ b \end{pmatrix} = \begin{pmatrix} \sum \sigma_{zi}^2 + \sum \sigma_{zi} \\ \sum \sigma_{zi} + n \end{pmatrix}^{-1} \begin{pmatrix} \sum \sigma_{zi} \sigma_{ii} \\ \sum \sigma_{ii} \end{pmatrix}$$

S.No	σ_{ii}	σ_{zi}	σ_{zi}^2	$\sigma_{ii} \cdot \sigma_{zi}$
1	8.01	1.4	1.96	11.214
2	11.14	2.5	6.25	27.85
3	13.82	3.1	9.61	42.842
4	18.40	5.2	27.04	95.68
5	22.3	5.5	30.25	122.65
Sum.	73.67	17.7	75.11	300.236

$$\begin{pmatrix} m \\ b \end{pmatrix} = \begin{pmatrix} 75.11 & 17.7 \\ 17.7 & 5 \end{pmatrix}^{-1} \begin{pmatrix} 300.236 \\ 73.67 \end{pmatrix}$$

Calc. Steps for matrix

Menu Setup $\rightarrow$ 4 $\leftarrow$

Define matrix 1 (A) $\leftarrow$

2x2

Value feed = $\leftarrow$

For calcn. we optn button Self (3)

Optn. button Self (matrix) A.

$$\begin{pmatrix} m \\ b \end{pmatrix} = \begin{pmatrix} 0.0830 & -0.284291 \\ -0.284291 & 1.20539 \end{pmatrix} \begin{pmatrix} 300.236 \\ 73.67 \end{pmatrix}$$

$$\begin{pmatrix} m \\ b \end{pmatrix} = \begin{pmatrix} 3.16769 \\ 3.5203 \end{pmatrix} \approx \begin{pmatrix} 3.17 \\ 3.52 \end{pmatrix}$$

Now from the Mohr - Coulomb formula

$$\sigma_1 = \frac{2c \cdot \cos \phi}{1 - \sin \phi} + \sigma_3 \left(\frac{1 + \sin \phi}{1 - \sin \phi} \right)$$

$$\Rightarrow m = 3.17 = \frac{1 + \sin \phi}{1 - \sin \phi}$$

$$\frac{3.17 + 1}{3.17 - 1} = \frac{1}{\sin \phi}$$

$$\sin \phi = \frac{3.17 - 1}{3.17 + 1} = \frac{2.17}{4.17} = 0.52038$$

$$\boxed{\phi = \sin^{-1}(0.52038) = 31.57^\circ} \text{ are}$$

$$\frac{2c \cos \phi}{1 - \sin \phi} = b = 3.52$$

$$\text{or } \frac{2c \cdot \cos(31.57)}{1 - \sin(31.57)} = 3.52$$

$$c = \frac{3.52 \times (1 - \sin 31.57)}{2 \cos(31.57)}$$

$$= 0.9842$$

$$\text{or } \boxed{C = 0.98 \text{ MPa}}$$

$$\boxed{\phi = 31.57^\circ}$$

Ans